

Assignments Continuous Integration Using Jenkins

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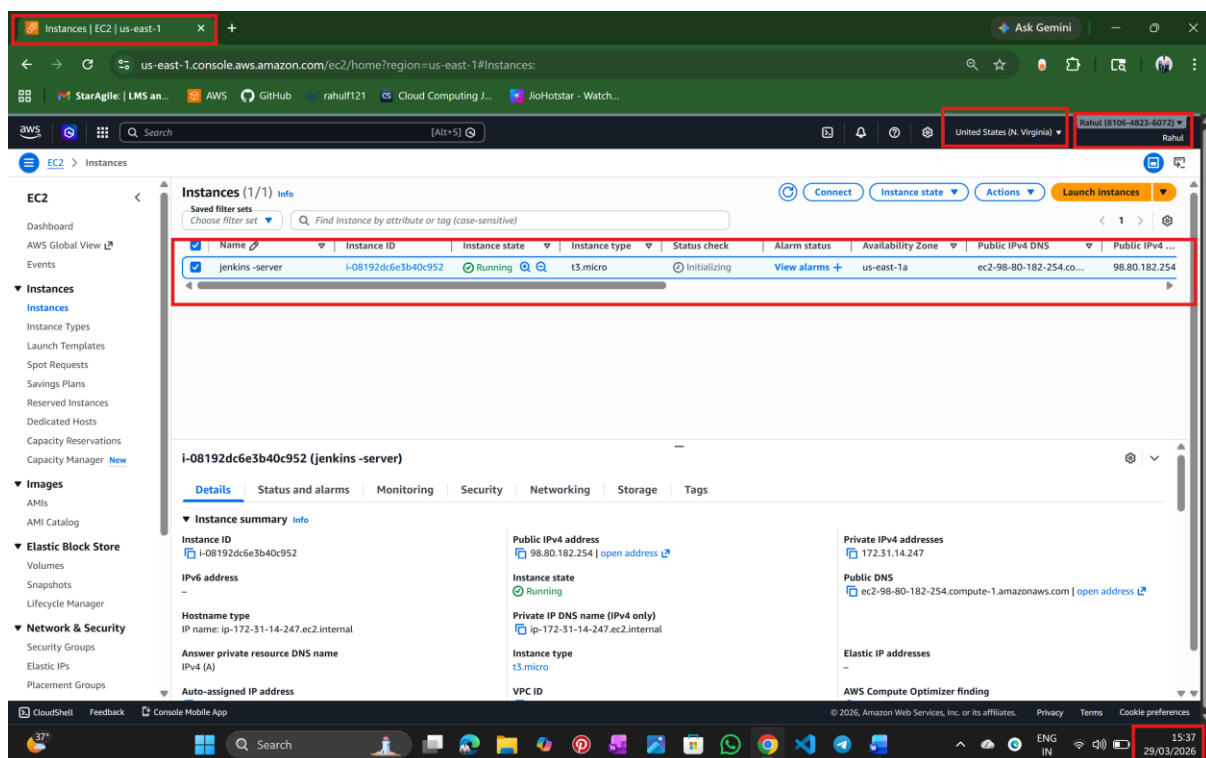
L2 - Create CICD Pipeline to Clone and Build Java Maven Web Application

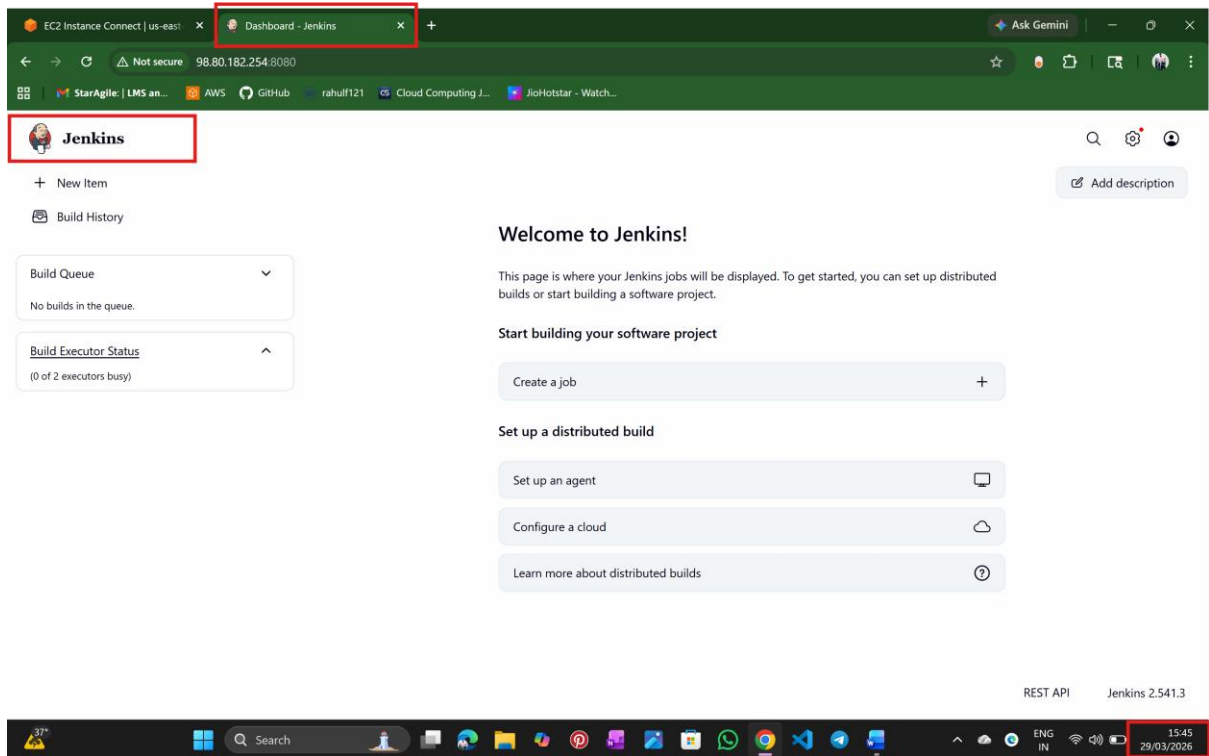
Step 1: Install Jenkins

→ Install Jenkins on your system or EC2 instance

→ Start Jenkins service

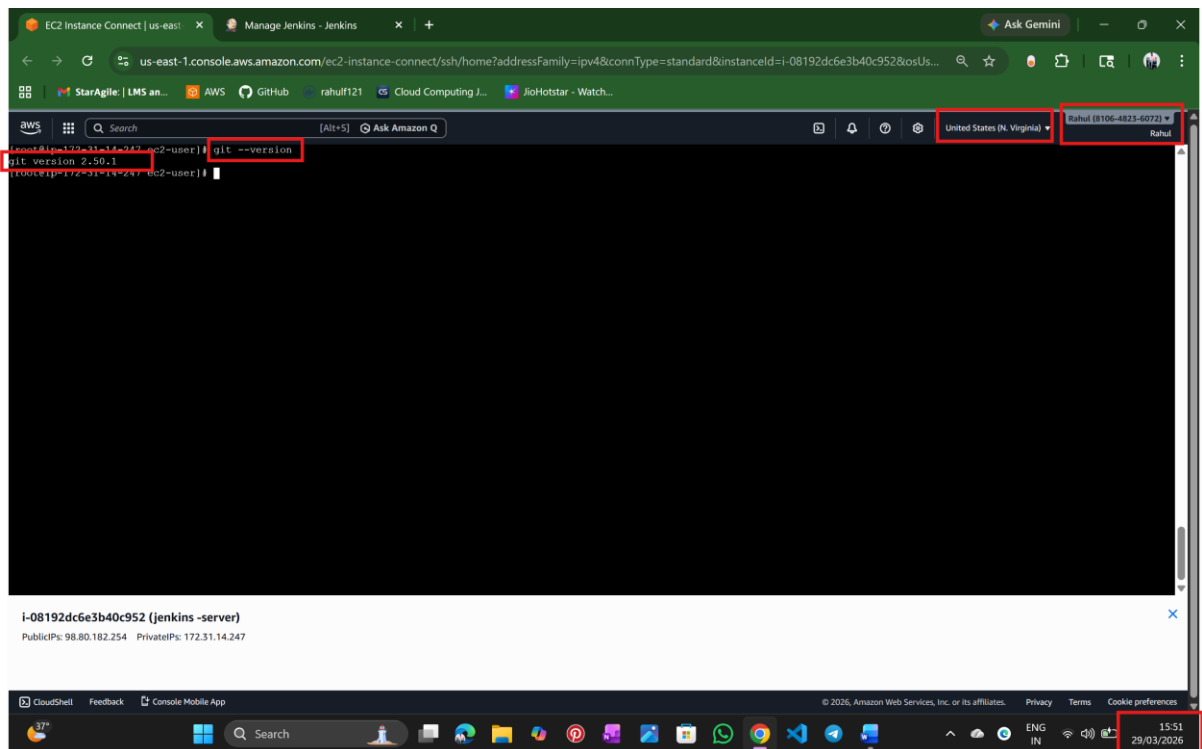
→ Open Jenkins in browser:





Step 2: Install Required Tools

→Git



Step 3: Configure Global Tools in Jenkins

Add:

→JDK

→Maven

The screenshot shows the Jenkins web interface at the URL `98.80.182.254:8080/manage/configureTools/`. The page title is "Tools - Manage Jenkins - Jenkins". Under the "JDK installations" section, there is a "+ Add JDK" button and a list of installed JDKs. One JDK is listed with the name "jdk17". The "Name" field is highlighted with a red box. Below the "Name" field, the "JAVA_HOME" field is highlighted with a red box and contains the value `/usr/lib/jvm/java-17-amazon-corretto`. A red error message is displayed below the "JAVA_HOME" field: `/usr/lib/jvm/java-17-amazon-corretto doesn't look like a JDK directory`. There is an "Install automatically" checkbox which is unchecked. At the bottom of the page, there are "Save" and "Apply" buttons. The system tray at the bottom right shows the time as 15:47 on 29/03/2026.

The screenshot shows the Jenkins web interface at the URL `98.80.182.254:8080/manage/configureTools/`. The page title is "Tools - Manage Jenkins - Jenkins". Under the "Maven" section, there is a "+ Add Maven" button and a list of installed Maven tools. One Maven tool is listed with the name "maven3". The "Name" field is highlighted with a red box. Below the "Name" field, the "Install automatically" checkbox is checked. Under the "Install from Apache" section, the "Version" field is highlighted with a red box and contains the value "3.8.8". At the bottom of the page, there are "Save" and "Apply" buttons. The system tray at the bottom right shows the time as 15:48 on 29/03/2026.

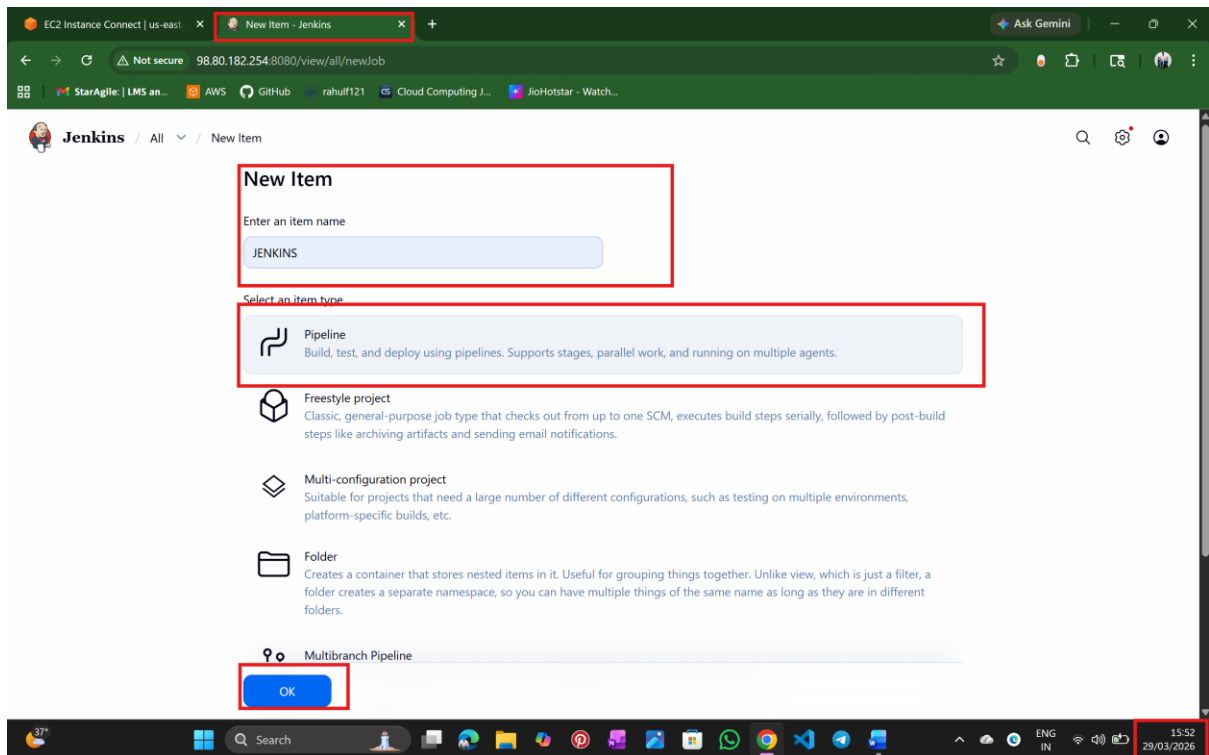
Step 4: Create Jenkins Pipeline Job

→ Click New Item

→ Enter job name: Jenkins

→ Select Pipeline

→ Click OK



Step 5: Configure Pipeline Script

The screenshot shows the Jenkins 'Configure' page for a pipeline. The 'Pipeline' tab is selected in the left sidebar. The 'Script' text area contains the following pipeline script:

```
1 pipeline {
2   agent any
3
4   stages {
5     stage('git checkout') {
6       steps {
7         git branch: 'main', url: 'https://github.com/raahulnalathati-21/java-tomcat-webapp.git'
8       }
9     }
10    stage('maven build') {
11      steps {
12        sh 'mvn clean package'
13      }
14    }
15  }
16 }
```

The 'Save' button is highlighted in the bottom right. The system tray at the bottom shows the date and time as 15:55 on 29/03/2026.

```
pipeline {
  agent any

  stages {
    stage('git checkout') {
      steps {
        git branch: 'main', url: 'https://github.com/raahulnalathati-21/java-tomcat-webapp.git'
      }
    }
    stage('maven build') {
      steps {
        sh 'mvn clean package'
      }
    }
  }
}
```

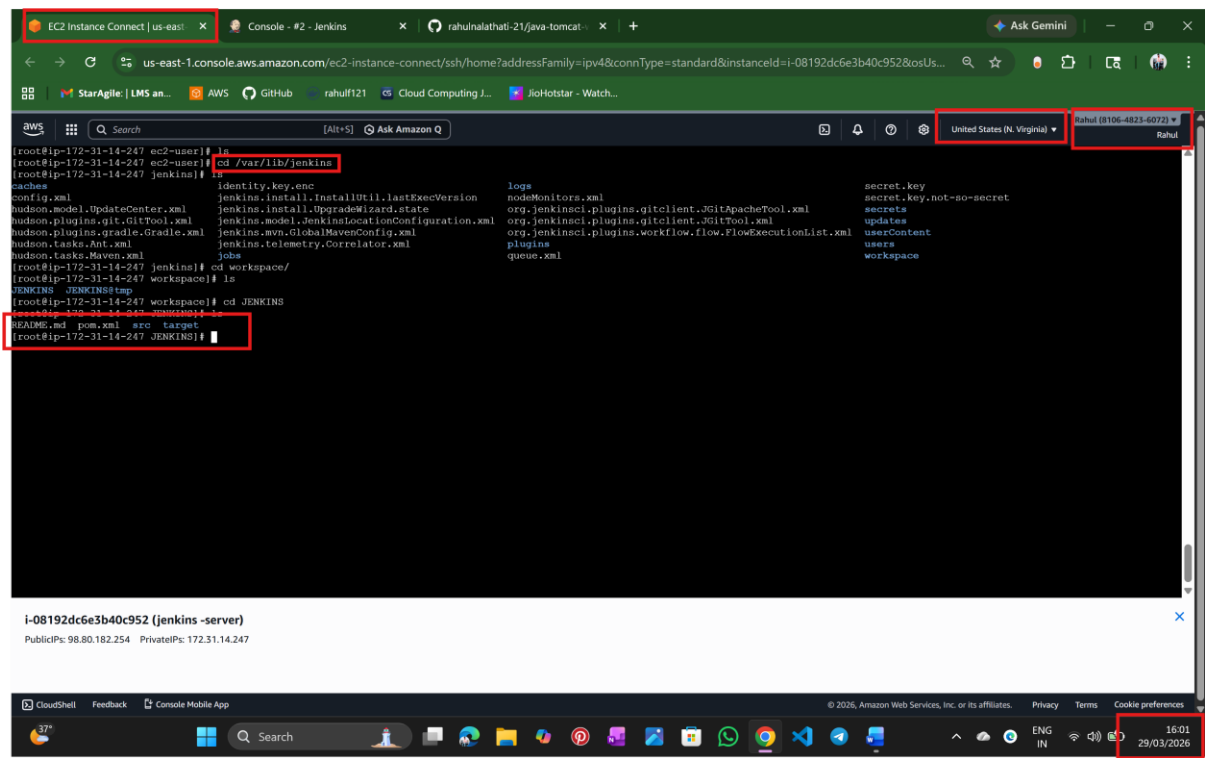
Step 6: Save and Build

The screenshot shows the Jenkins web interface. The 'JENKINS' tab is highlighted with a red box. The 'Builds' section shows a list of builds, with build #2 (10:28 AM) highlighted by a red box. The system tray at the bottom shows the time as 15:58 on 29/03/2026, also highlighted by a red box.

Step 7: Verify Build Output

The screenshot shows the Jenkins web interface with the 'Console Output' tab selected and highlighted by a red box. The console output displays the build process, including the command 'Running on Jenkins in /var/lib/jenkins/workspace/JENKINS' which is highlighted by a red box. The system tray at the bottom shows the time as 15:59 on 29/03/2026, also highlighted by a red box.

Step 8: Check Generated Artifact



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